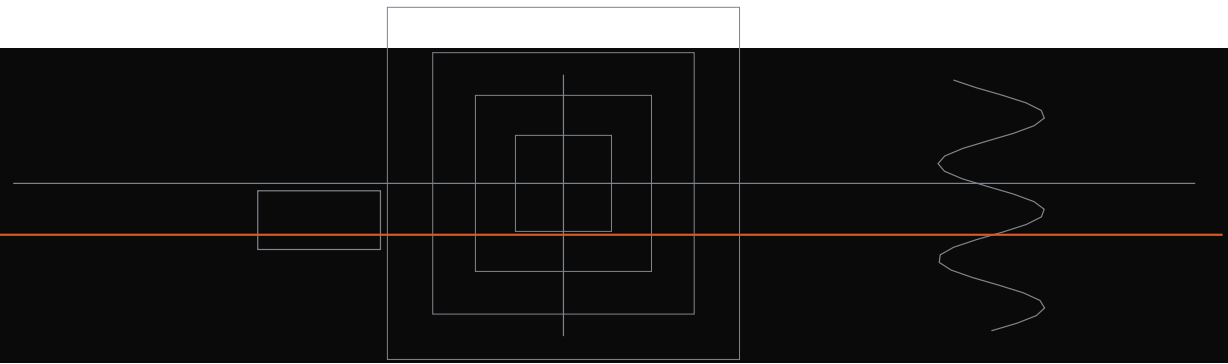


BILL OF MATERIALS

# Quadruped I

Rev. 2026-04-30 67 items



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**BOM**

Product/Project: Quadrupe I  
Revision date: 2026-04-30

<https://www.printables.com/model/1443108-quadrupe-i>

**LEG LINKAGE MECHANISM**

11 items

|    | Description                   | Qty        | AliExpress                      | Amazon (Geniuslink)         | Notes                                                                           |
|----|-------------------------------|------------|---------------------------------|-----------------------------|---------------------------------------------------------------------------------|
| >> | Servo Motors                  | <b>x12</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 180 degrees, 35kg (Approx 49.5x20,2x40mm)                                       |
| >> | Raspberry Pi 4 Model B        | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 8gb Ram                                                                         |
| >> | LiPo Battery                  | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 4000mAh 4S 45~90C Lipo Pack w/XT-90 (You can change to the XT-90 connect later) |
| >> | IMU                           | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Integrated Measuring Unit                                                       |
| >> | Raspberry Pi Breakout Board   | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Made by Dfrobot                                                                 |
| >> | Fan                           | <b>x2</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                                                               |
| >> | Servo Driver PCA9685          | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | To control servo motors, via I2C interface                                      |
| >> | Arduino Nano                  | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For IMU, Battery sensing, LCD screen                                            |
| >> | DC to DC (Buck) converter 20A | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Adjustable, set to ~7V in Dingo                                                 |
| >> | Power Distribution PCB        | <b>1</b>   | <a href="#">Order PCB HERE</a>  |                             | Make sure to chose PCB + Assembly                                               |
| >> | Orange Acrylic                | <b>1</b>   | -                               | -                           | Adding STEP + DXF in case you cant print this.                                  |

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**BODY CONSTRUCTION**

4 items

|    | Description                  | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                       |
|----|------------------------------|-----|---------------------------------|-----------------------------|-----------------------------|
| >> | LCD Screen                   | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 1.47 inch                   |
| >> | DC to DC (buck) converter 3A | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For fans                    |
| >> | Emergency Stop Switch        | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 3a at 12-220V rated         |
| >> | 32GB MicroSD card            | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 32GB is maximum Pi supports |

|            |                                                          |                                                                                                                       |
|------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
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|------------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|

| <div> <b>COMPUTING</b> <span>7 items</span> </div> |                              |            |                                 |                             |                                               |
|----------------------------------------------------|------------------------------|------------|---------------------------------|-----------------------------|-----------------------------------------------|
|                                                    | Description                  | Qty        | AliExpress                      | Amazon (Geniuslink)         | Notes                                         |
| >>                                                 | Bearing                      | <b>x4</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 6705-2RS 25X32X4                              |
| >>                                                 | Thrust Bearings (legs)       | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | F4-10M 4x10x4mm                               |
| >>                                                 | Bearing                      | <b>x20</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 684ZZ Mini Deep Groove Ball Bearing 4x9x4mm   |
| >>                                                 | Servo horns                  | <b>x4</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | RC Car Steering Servo Horn Arm 25T Teeth Long |
| >>                                                 | DC to DC (Buck) converter 4A | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For all components except servos and fans     |
| >>                                                 | RasbPi Cooling System        | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                             |
| >>                                                 | Arduino Nano Breakout Board  | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                             |

|     |                             |                                                                                                                       |
|-----|-----------------------------|-----------------------------------------------------------------------------------------------------------------------|
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|     | Revision date: 2026-04-30   |                                                                                                                       |

| POWER SUPPLY AND MANAGEMENT 4 items |                 |     |                                 |                             |                                            |
|-------------------------------------|-----------------|-----|---------------------------------|-----------------------------|--------------------------------------------|
|                                     | Description     | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                                      |
| >>                                  | Terminal Blocks | x8  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 1935776                                    |
| >>                                  | Heatsinks       | x4  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 6mm, black. Cut in half for each leg motor |
| >>                                  | GPIO Cable      | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | To connect Pi to its breakout board        |
| >>                                  | Servo Discs     | x10 | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | with screws.                               |

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**PCB PARTS**

10 items

|    | Description           | Qty       | AliExpress                      | Amazon (Geniuslink)         | Notes                                             |
|----|-----------------------|-----------|---------------------------------|-----------------------------|---------------------------------------------------|
| >> | Magnets               | <b>x8</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Used to hold fans in place                        |
| >> | Rod ends,Long         | <b>x4</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | S 3mm L30mm M3                                    |
| >> | Rod long              | <b>x4</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 90mm long (cut if needed)                         |
| >> | Rod ends Short        | <b>x4</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | N 3mm L20mm M3                                    |
| >> | Rod short             | <b>x2</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 16mm long (cut if needed)                         |
| >> | Logic Level Converter | <b>1</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For adjusting voltage between the pi and the nano |
| >> | M3 threaded inserts   | <b>1</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Pack of 100, M3xD4.6xL4.0                         |
| >> | XT60H Connector       | <b>1</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | To connect LiPo to the robot                      |
| >> | On/off Switch         | <b>1</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 20A at 14VDC rated                                |
| >> | Spring                | <b>1</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For battery latch mechanism. 35mm long x 7mm OD   |

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**ELECTRICAL**

9 items

|    | Description                        | Qty        | AliExpress                      | Amazon (Geniuslink)         | Notes                                                   |
|----|------------------------------------|------------|---------------------------------|-----------------------------|---------------------------------------------------------|
| >> | Male Connector (for peripherals)   | <b>x12</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | B2B-PH-K-S(LF)(SN)                                      |
| >> | Digital Voltmeter                  | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | DC 2.5-30V, used to measure battery voltage             |
| >> | Socket for Connectors              | <b>x25</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | SPH-002T-P0.5S                                          |
| >> | Threaded rods, long                | <b>x5</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 x 110mm                                              |
| >> | Female Connector (for peripherals) | <b>x12</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | PHR-2                                                   |
| >> | Capacitor                          | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 2200uF 16VDC Electrolytic RB Capacitor ,For servo board |
| >> | Threaded rods,short                | <b>x5</b>  | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | M3 x 25mm                                               |
| >> | Reverse Polarity Protection Diode  | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | B160-13-F                                               |
| >> | 47k Resistor                       | <b>1</b>   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | CRGCQ1206F47K                                           |

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**MISC**

9 items

|    | Description         | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes                                                                               |
|----|---------------------|-----|---------------------------------|-----------------------------|-------------------------------------------------------------------------------------|
| >> | 10k resistor        | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | CRGCQ0805F10K                                                                       |
| >> | PS4 Controller      | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                                                                   |
| >> | Power Delivery Wire | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Minimum 12AWG, for wires expected to carry current above 2A and/or voltage above 5V |
| >> | Connection Wire     | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | 22-30AWG, recommend solid core, used for all other wires                            |
| >> | Ferrules            | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Optional, if used need approx 20 across robot                                       |
| >> | Heat Shrink         | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Only small amount required, for covering exposed solid-core wires                   |
| >> | 100k resistors      | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | For dividing the voltage at the buck converter so that it is readable by the nano   |
| >> | M3 Nuts             | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                                                                   |
| >> | M3 Bolts            | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -                                                                                   |



|            |                                                           |                                                                                                                         |
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|------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|

| <div>FILAMENTS</div> <div>4 items</div> |                                 |          |                                 |                             |                                                    |
|-----------------------------------------|---------------------------------|----------|---------------------------------|-----------------------------|----------------------------------------------------|
|                                         | Description                     | Qty      | AliExpress                      | Amazon (Geniuslink)         | Notes                                              |
| >>                                      | Black ABS/ASA/PETG&PC Filament  | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Approx 500g required                               |
| >>                                      | Yellow ABS/ASA/PETG&PC Filament | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Approx 500g required                               |
| >>                                      | Black PLA Filament              | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Approx 750g required (includes 500g for the stand) |
| >>                                      | White TPU Filament              | <b>1</b> | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | Approx 50g required, for feet                      |

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**TOOLS**

9 items

|    | Description               | Qty | AliExpress                      | Amazon (Geniuslink)         | Notes |
|----|---------------------------|-----|---------------------------------|-----------------------------|-------|
| >> | Crimp Tool                | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Hex Key set               | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Soldering Iron            | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Wire cutter               | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Wire stripper             | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Precision Screwdriver Set | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Heat insert set           | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Heat insert tip - set     | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |
| >> | Soldering Iron TS101      | 1   | <a href="#">AliExpress Link</a> | <a href="#">Amazon Link</a> | -     |